Topic : Clock

Pre-Assessment:

A watch which gains uniformly, is 5mins slow at 8 o'clock in the morning on Sunday and it is 5min 48 sec fast at 8 P.M on the following Sunday. When was it correct?

____

Concept:

Type1:

An accurate clock shows 8 o'clock in the morning. Through how degrees may will the hour hand rotate when the clock shows 2 o'clock in the afternoon?

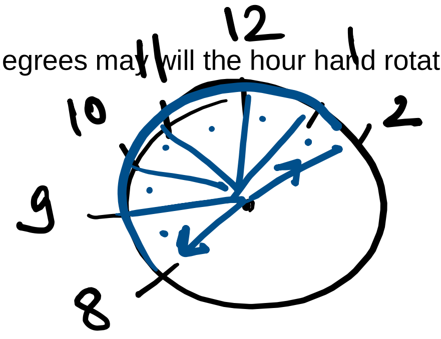
A. 144°

B. 150°

C. 168°

[✓ D.] 180°

$$6 \times 30^\circ = 180^\circ$$



6:20

Imp.

A clock is started at noon. By 10 minutes past 5, the hour hand has turned through:

A. 145°

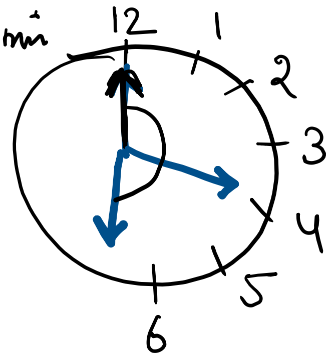
B. 150°

[✓ C.] 155°

D. 160°

$$Q = 30^\circ \times 5 + \frac{1}{2} \times 10\,\text{min}$$

$$\boxed{Q = 155^\circ}$$



6:20

Hour hand

$$6 \times 30^\circ + \frac{1}{2} \times 20$$

$$180 + 10 = \underline{\underline{190^\circ}}$$

Type2:

At what time, in minutes, between 3 o'clock and 4 o'clock, both the needles will coincide each other?

- **A.** $5\frac{1}{11}''$
- **B.** $12\frac{4}{11}''$
- **C.** $13\frac{4}{11}''$
- **D.** $16\frac{4}{11}''$

Type3:

At what angle the hands of a clock are inclined at 15 minutes past 5?

- **A.** $58\frac{1}{2}^\circ$
- **B.** 64°
- **C.** $67\frac{1}{2}^\circ$
- **D.** $72\frac{1}{2}^\circ$

At 3:40, the hour hand and the minute hand of a clock form an angle of:

- **A.** 120°
- **B.** 125°
- **C.** 130°
- **D.** 135°

$$\theta = \left|30 \times 3 - \frac{11}{2} \times 40\right|$$

Type 4:

The angle between the minute hand and the hour hand of a clock when the time is 4.20, is:

- **A.** 0°
- [✓ **B.** 10°]

C. 5°

D. 20°

Type 5:

$(\boxed{7}-8)$

At what time between 7 and 8 o'clock will the hands of a clock be in the same straight line but, not together?

A. 5 min. past 7

B. $5\frac{2}{11}$ min. past 7

[✓ C. $5\frac{5}{11}$ min. past]

D. $5\frac{3}{11}$ min. past 7

Ans

$\theta = \left|30H - \frac{11}{2}m\right|$

$180 = 30 \times 7 - \frac{11}{2} \times M$

$\frac{11}{2}m = 30 \Rightarrow m = \frac{60}{11} = 5\frac{5}{11}$ min

Question

At what time between 4 and 5 clock will the hands of a clock be at right angle?

$M > H$

4-5

$38\frac{3}{11}$ minutes past 4

$38\frac{9}{11}$ minutes past 4

[✓ $38\frac{2}{11}$ minutes past 4]

$38\frac{5}{13}$ minutes past 4

$90^\circ = 30 \times 4 - \frac{11}{2}m$

$90 = 120 - \frac{11}{2}m$

$\frac{11}{2}m = 30 \Rightarrow m = \frac{60}{11}$

$\boxed{m = 5\frac{5}{11}}$

$\theta = \frac{11}{2}m - 30H$ 4

$90 = \frac{11}{2}M - 120$

$\frac{420}{11} = 38\frac{2}{11}$

Question:

At what time between 3 and 4 will the hands of a clock 4 minutes apart?

1. 12 min past 3 & $20\frac{8}{11}$ past 3
2. 13 min past 3 & $25\frac{8}{11}$ past 3
3. 12 min past 3 & $20\frac{8}{11}$ past 3
4. 13 min past 3 & $19\frac{8}{11}$ past 3

Hint = $\theta = 24^\circ$

$24 = 30 \times 3 - \frac{11}{2}m$

$\theta = 12^\circ$

$24 = \frac{11}{2}m - 30 \times 3$

$\theta = 20\frac{8}{11}$

Type 6:

Question: [Unitary method]

Mohit saw the clock when it was set right at 8 a.m. The clock gains 5mins in an hour. What time will it shows at 8p.m at the same day?

9 pm Ans

9 p.m Ans

1 hr — + 5min

12 hr → + 60min

Mohit observe that a watch loses 5 seconds in one hour which was right set at 7 a.m. What time will it show at 2p.m on the same day?

$\boxed{1:59:25}$ Pm

1 hr → (-) 5 sec

7 hr → 35 sec

Question:

A watch is showing right time at 9 P.M. This watch gains 10 minutes in every 24 Hours. What will be the time shown next day by the watch when the correct time is 2 A.M?

2:2:05 Am

24 Hr = +10 mins

$$1\text{ Hr }(5\text{ hrs}) = \frac{\cancel{10}\,5}{\cancel{24}\,12} \times 5 = \frac{25}{12}$$

$$2\frac{1}{12} = +2\text{ mins }5\text{ sec}$$

Type 7:

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Rise

$24 \times 7 = 168$

$+12 = 180$

$10\frac{\cancel{48}\,4}{\cancel{60}\,5}$

Sun — 8' am ⟶ 5 mis (−)

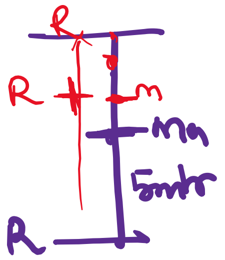
Sun — 8' pm ⟶ 5 mi 48 (+)

180 Hr ⟶ 10 mis 48 sec

$\frac{54}{5}$ mins is gained in 180 Hr

1 mins ⟶ $\frac{180 \times 5}{54}$ Hr

5 mis ⟶ $\frac{\overset{10}{\cancel{180}} \times 5 \times 5}{\underset{3}{\cancel{54}}} = \frac{250}{3} \Rightarrow 83\frac{1}{3}$ Hrs



Sunday 8am ⟶ 3 days + 11hr 20mins
(72 Hr)

8am Wed

7:20 pm on Wednesday